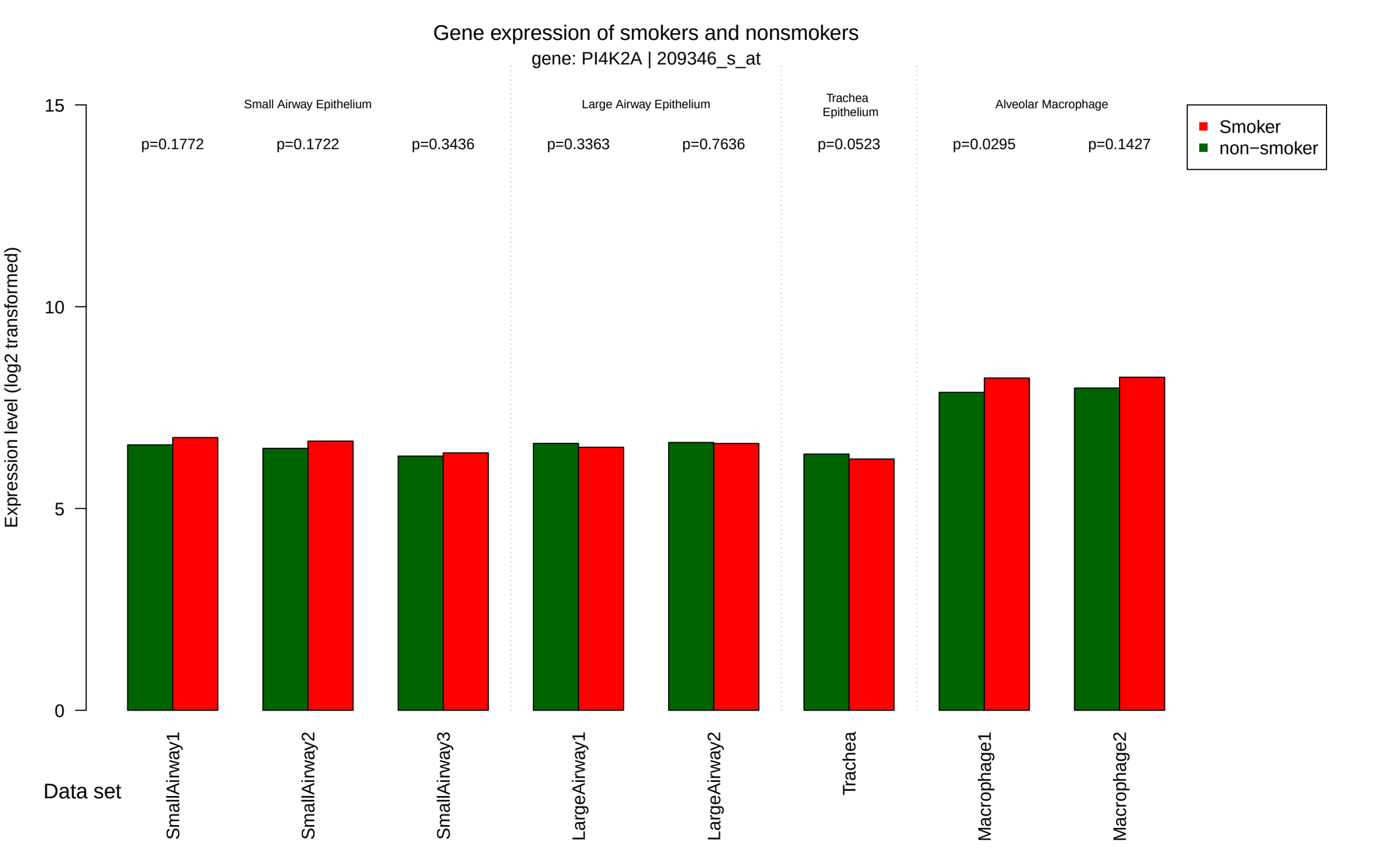
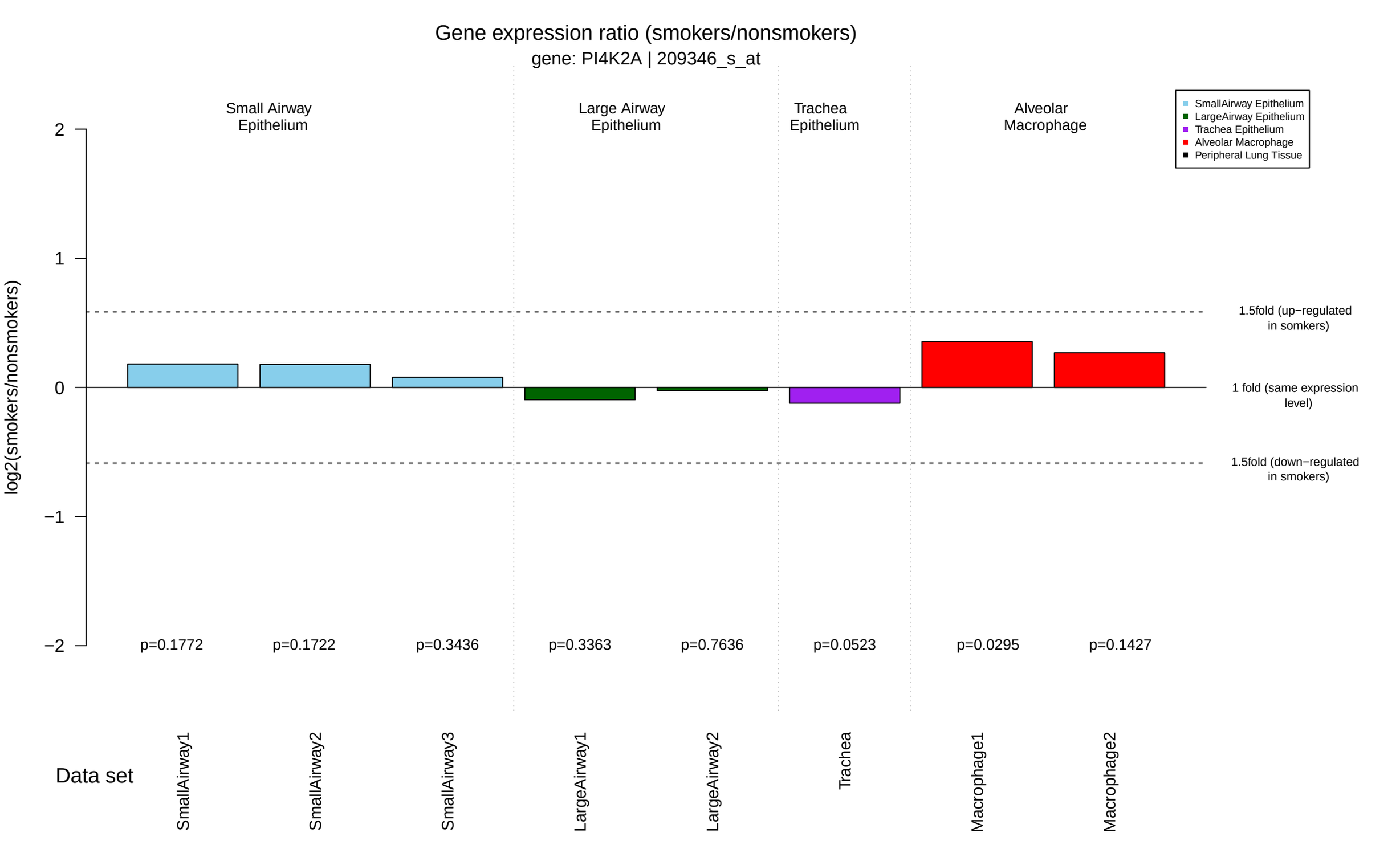
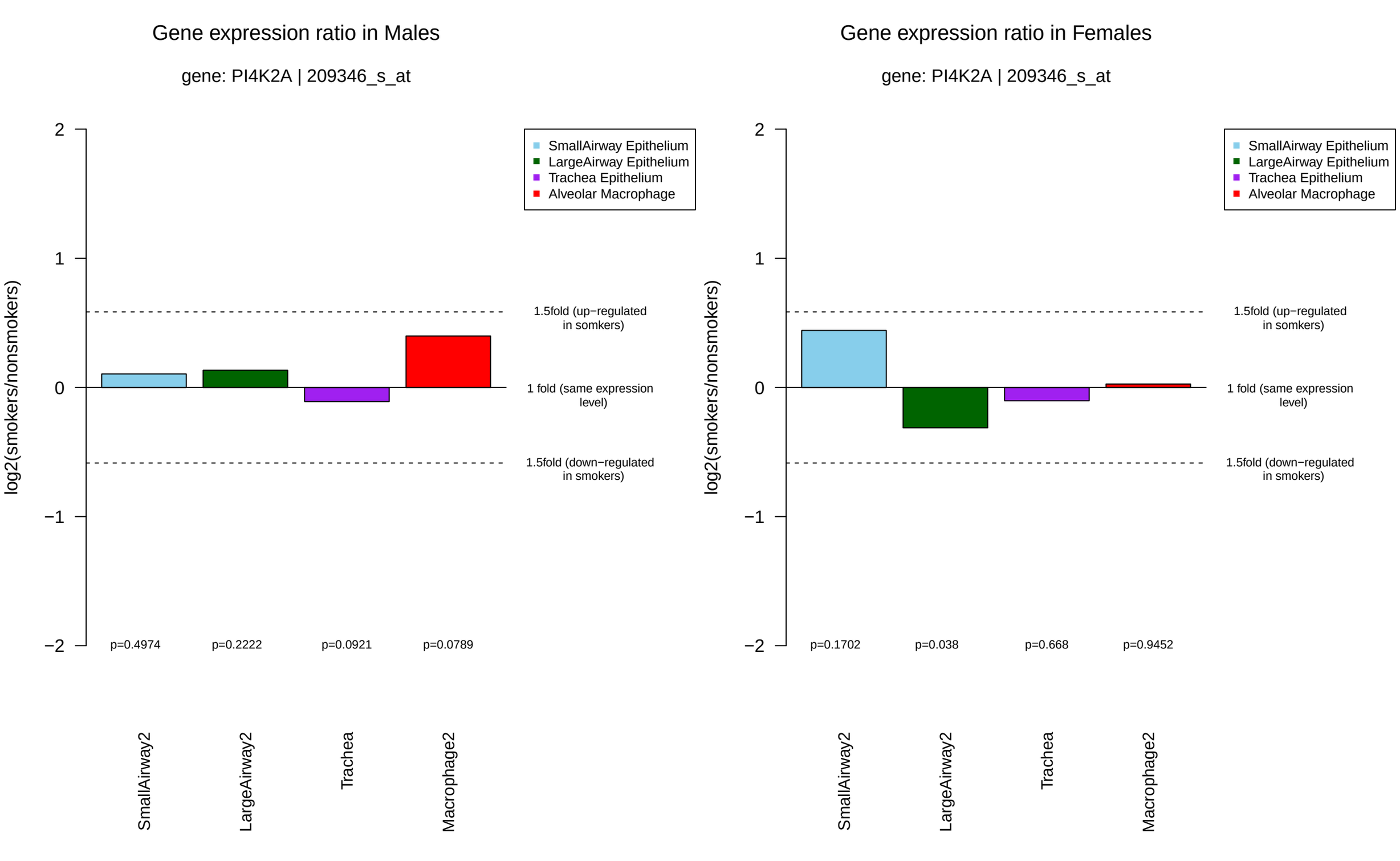
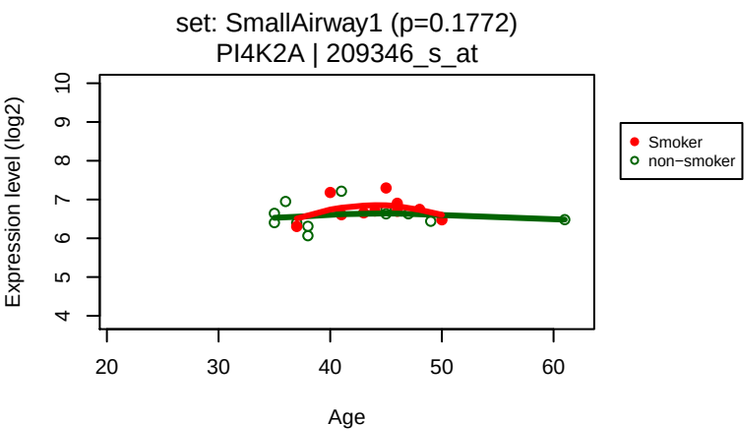
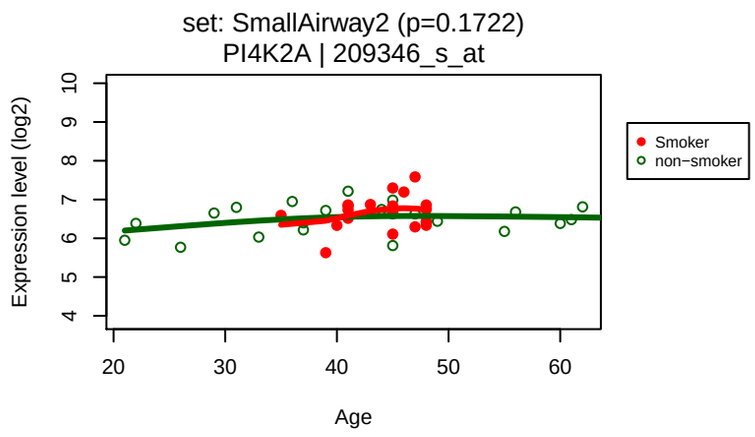
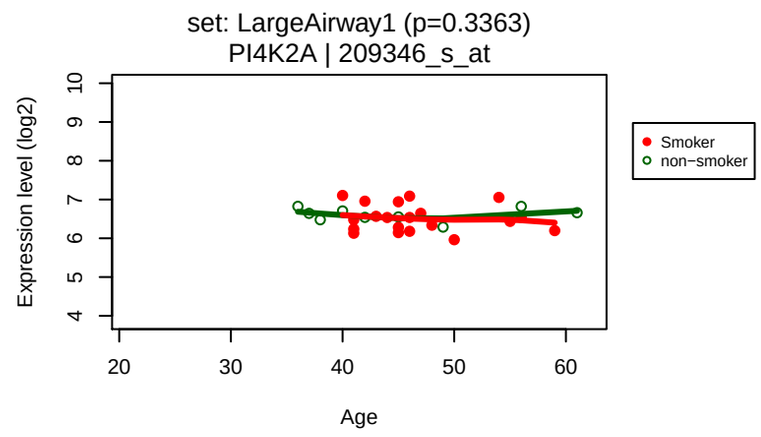

Gene expression of smokers and nonsmokers
gene: PI4K2A | 209346_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1772
p=0.1722
p=0.3436
p=0.3363
p=0.7636
p=0.0523
p=0.0295
p=0.1427
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

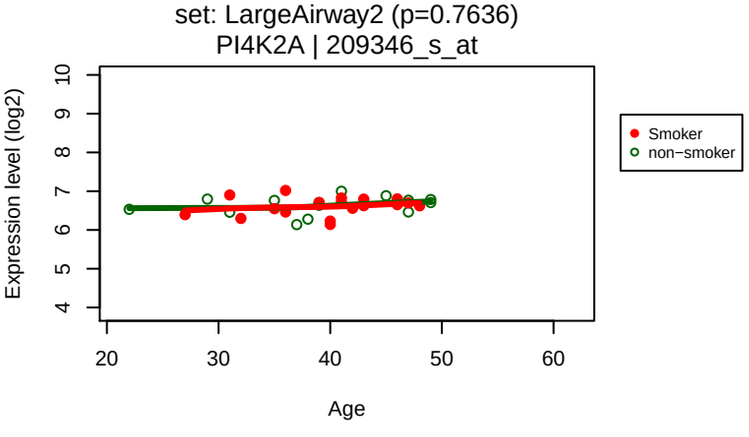

Gene expression ratio (smokers/nonsmokers)
gene: PI4K2A | 209346_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1772
p=0.1722
p=0.3436
p=0.3363
p=0.7636
p=0.0523
p=0.0295
p=0.1427
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

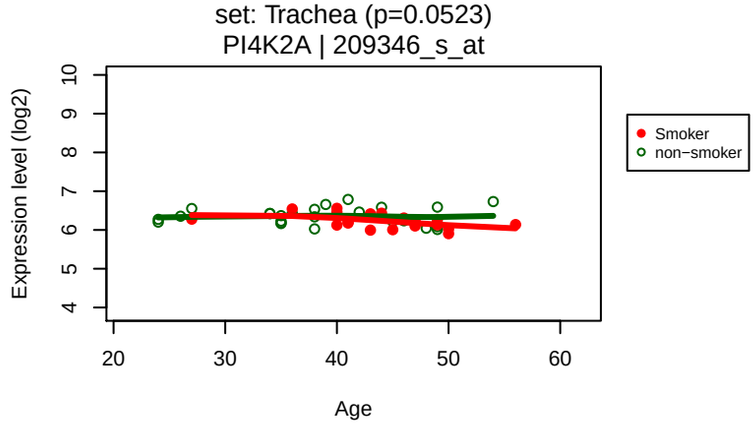

Gene expression ratio in Males
gene: PI4K2A | 209346_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4974
p=0.2222
p=0.0921
p=0.0789
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PI4K2A | 209346_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1702
p=0.038
p=0.668
p=0.9452
SmallAirway2
LargeAirway2
Trachea
Macrophage2

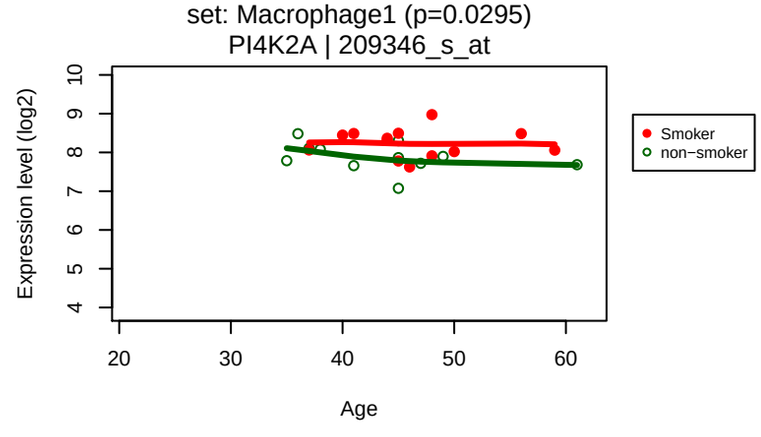

set: SmallAirway1 (p=0.1772)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker

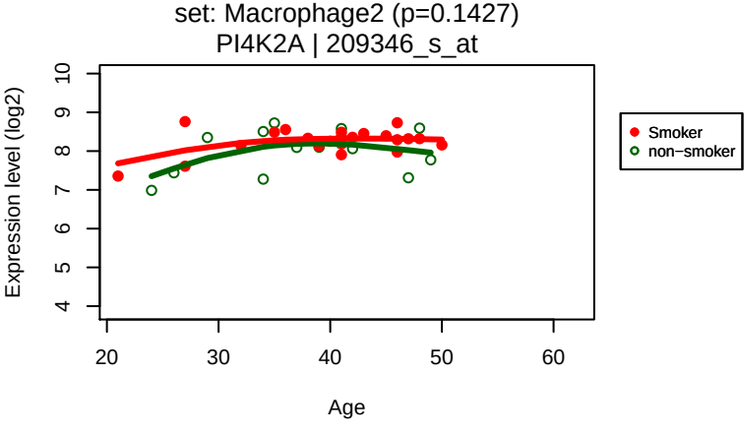

set: SmallAirway2 (p=0.1722)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker

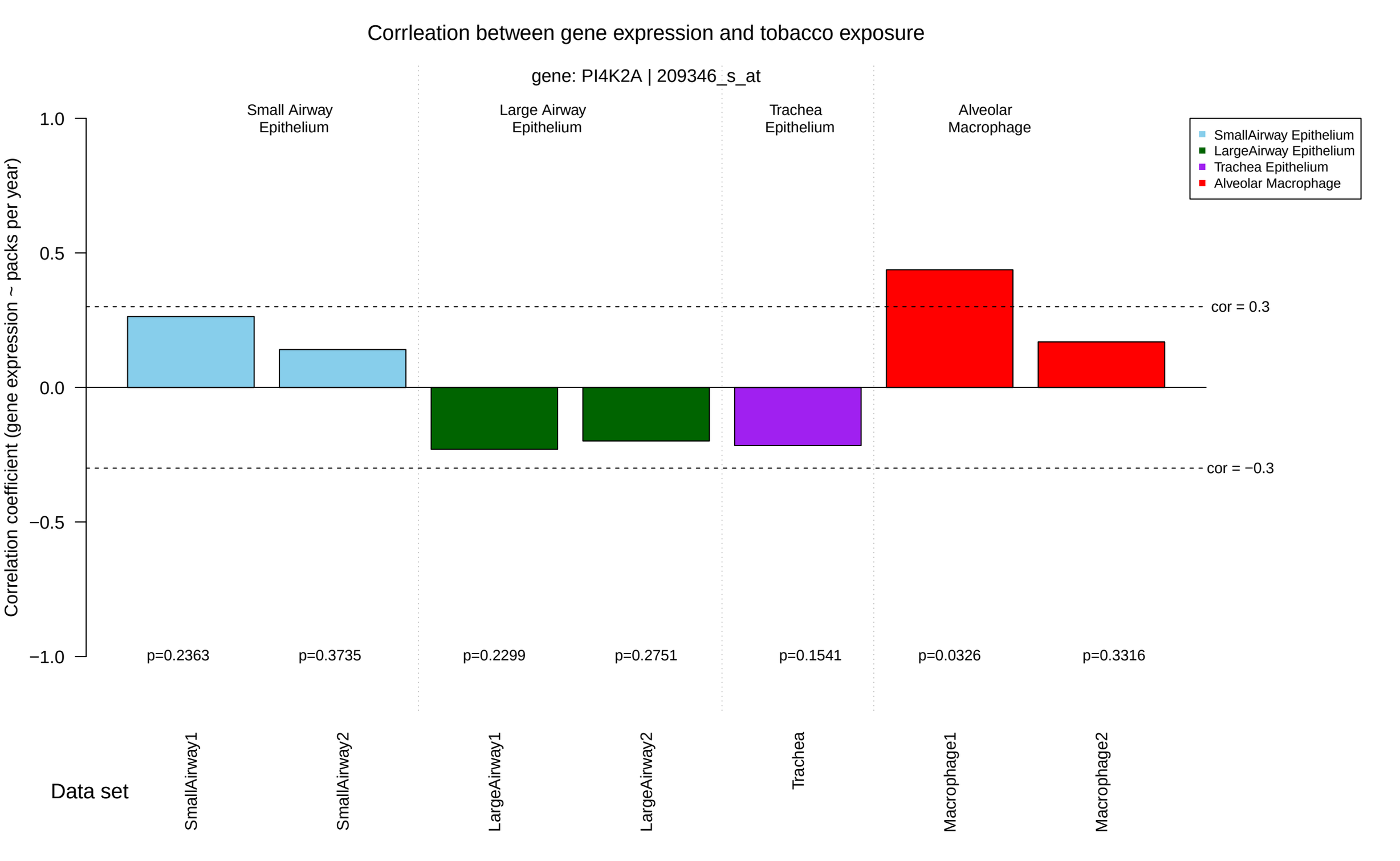

set: LargeAirway1 (p=0.3363)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7636)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0523)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0295)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1427)
PI4K2A | 209346_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PI4K2A | 209346_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2363
p=0.3735
p=0.2299
p=0.2751
p=0.1541
p=0.0326
p=0.3316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2